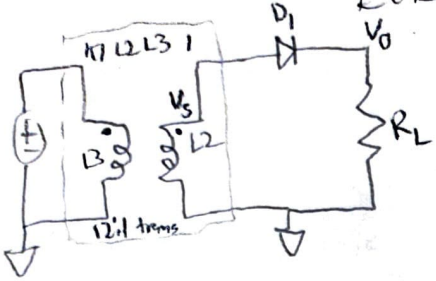
# ECEN 164 LAB 4 PreLab

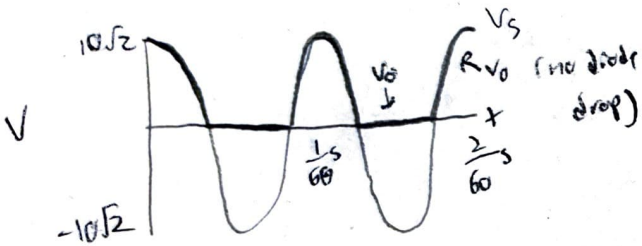
Dylan Tremblay




← probably supposed to be 500

a. $v_i(t) = 120\sqrt{2}\cos(2\pi 60t)$

b. $v_s(t) = 10\sqrt{2}\cos(2\pi 60t)$




(no diode drop)

c. $V_{avg} = \frac{V_{peak}}{\pi} = \boxed{\frac{10\sqrt{2}}{\pi} V}$

d. $V_{RMS} = \frac{10\sqrt{2}}{2} = 5\sqrt{2}$

$P_{avg} = \frac{(5\sqrt{2})^2}{500} = \boxed{.1 W}$

e. $S = \frac{10\sqrt{2}}{\sqrt{2}}\left(\frac{5\sqrt{2}}{500}\right) = \frac{\sqrt{2}}{10} W$

$PF = \frac{.1}{\sqrt{2}/10} = \boxed{\frac{\sqrt{2}}{2}}$

f.

DC component (n=0)

60Hz (n=1)

120, 180, 240...

(n=2, 3, 4...)

f.

DC comp (n=0)

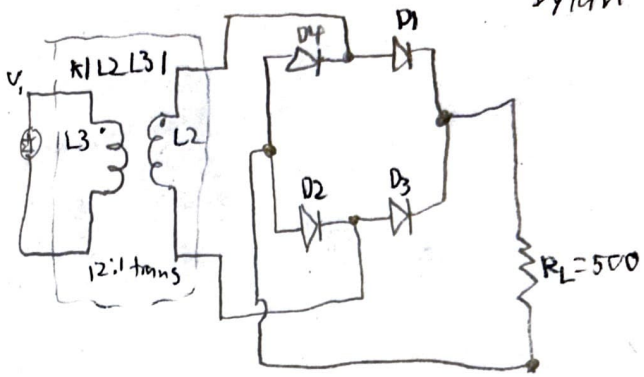
120Hz (n=2)

240, 480, 720...

(n=4, 6, 8...)

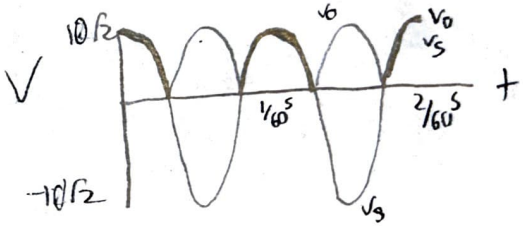
This is the behavior we expect. Why? B/c that's what the definitions say.

:)




a. $v_i(t) = 120\sqrt{2}\cos(2\pi 60t)$

b. $V_s(t) = 10\sqrt{2}\cos(2\pi 60t)$




c. $V_{avg} = \frac{2V_{peak}}{\pi} = \boxed{\frac{20\sqrt{2}}{\pi} V}$

d. $V_{RMS} = \frac{10\sqrt{2}}{\sqrt{2}} = 10$

$P_{avg} = \frac{10^2}{500} = \boxed{.2 W}$

e. $\frac{10}{500} = \frac{1}{50} A$

$S = 10\left(\frac{1}{50}\right) = .2$

$\boxed{PF = \frac{.2}{.2} = 1}$